Minh Triet Pham

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Education

The University of Melbourne — Melbourne, Australia
Bachelor of Science in Data Science, First Class Honours — *Expected Nov. 2026*

- **Relevant Coursework** (**current*): *Artificial Intelligence, *Machine Learning, Algorithms & Data Structures, Databases, Object-Oriented Programming, Linear Algebra, Probability, Statistics

Experience

Undergraduate Research Fellow — Jan. 2026 – Present
The University of Melbourne - School of Mathematics and Statistics — *Melbourne, Australia*

- **Research Topic**: Universality in Network Embeddings | **Supervisor**: Dr. Ian Gallagher
- Analyzing and comparing network embedding algorithms to identify shared structural patterns and representation differences across diverse graph datasets

AI Engineer — Jul. 2025 – Oct. 2025
AI @ DSCubed - autonomous AI systems development — *Melbourne, Australia*

- Minimized API consumption costs by 25% through LLM prompt engineering and token optimization using LangChain across production agent pipelines
- Built Discord assistant agent integrating Notion and Discord APIs to automate team workflows and task management, boosting productivity by 15%
- Accelerated CI/CD deployment cycles by 20% via containerized Docker workflows and cut FastAPI endpoint latency by 10% through architectural refactoring

AI Engineer Intern - Computer Vision — May 2025 – Jul. 2025
Elcom - computer vision technology solutions — *Hybrid*

- Developed automated license plate recognition system detecting car numbers and vehicle attributes for 100,000+ images in high-speed road environments using YOLO and TensorFlow
- Implemented face recognition model for automated employee attendance system using OpenCV, replacing manual check-in processes
- Crawled, curated, and preprocessed 1,000,000+ images from diverse sources to build high-quality training datasets for computer vision models

Publication

Minh Triet Pham, Quynh Chi Dang, Nhat Tan Le. "Deep Attention-based Sequential Ensemble Learning for BLE-Based Indoor Localization in Care Facilities." *8th International Conference on Activity and Behavior Computing (ABC 2026), IEEE Technically Co-sponsored. Muroran & Hakodate, Japan, Mar. 2026.* Accepted · Revised version submitted · View Paper

Projects

SmartStudy - AI-Powered Learning Platform | *Python, FastAPI, LangChain, React, MongoDB, Docker, Render* — Link

- Developed full-stack intelligent study platform with AI tutoring conversations and personalized checklist tracking, deployed live with 50+ real trial users
- Architected full-stack application with React frontend, MongoDB for persistent storage, FastAPI backend, containerized with Docker, and deployed on Render
- Engineered LangChain integration as the LLM brain powering real-time tutoring responses across multi-turn conversations

RAG - Migration Act Assistant | *Python, Flask, LangChain, Render* — Link

- A RAG-powered legal AI chatbot retrieving answers from 1,000+ preprocessed Migration Act pages, served via Flask and deployed on Render
- Built end-to-end RAG pipeline with ChromaDB vector storage and LangChain as the LLM brain, enabling accurate legal query responses
- Designed hierarchical tree-based search reducing embedding lookups and query time by 10,000x over naive approaches

Technical Skills

Programming Languages: Python, Java, C/C++, JavaScript (React), SQL (MySQL), NoSQL (MongoDB)
Developer Tools: Git (GitHub/GitLab), Docker
Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, Pandas, NumPy, Matplotlib, BeautifulSoup